

Q-ViT: Fully Differentiable Quantization for Vision Transformer*

Zhexin Li^{1,3}

Tong Yang²

Peisong Wang¹

Jian Cheng^{1,†}

¹Institute of Automation, Chinese Academy of Sciences ²Megvii Technology

³School of Artificial Intelligence, [University of Chinese Academy of Sciences](https://github.com/zhexinli/Q-ViT-DeiT)

{zhexin.li, peisong.wang, jcheng}@nlpr.ia.ac.cn, yangtong@megvii.com

Abstract

In this paper, we propose a fully differentiable quantization method for vision transformer (ViT) named as *Q-ViT*, in which both of the quantization scales and bit-widths are learnable parameters. Specifically, based on our observation that heads in ViT display different quantization robustness, we leverage head-wise bit-width to squeeze the size of *Q-ViT* while preserving performance. In addition, we propose a novel technique named switchable scale to resolve the convergence problem in the joint training of quantization scales and bit-widths. In this way, *Q-ViT* pushes the limits of ViT quantization to 3-bit without heavy performance drop. Moreover, we analyze the quantization robustness of every architecture component of ViT and show that the Multi-head Self-Attention (MSA) and the Gaussian Error Linear Units (GELU) are the key aspects for ViT quantization. This study provides some insights for further research about ViT quantization. Extensive experiments on different ViT models, such as DeiT and Swin Transformer show the effectiveness of our quantization method. In particular, our method outperforms the state-of-the-art uniform quantization method by 1.5% on DeiT-Tiny.

1. Introduction

Recently, the vision transformer (ViT) has achieved great success in computer vision, such as image classification [9, 20, 29], object detection [7], and instance segmentation [20]. However, the vision transformer suffers from heavy computation and intensive memory cost. Thus, it is impractical to apply large vision transformer models to edge devices. To resolve these problems, many compression technologies, such as structured pruning [24, 25, 27], quantization [21], linear transformer [26], and customized acceleration [5], are applied to vision transformer. Although quan-

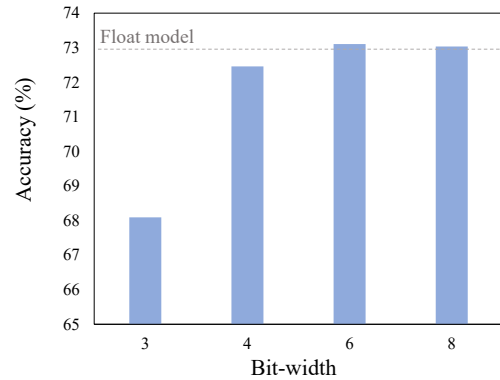


Figure 1. Quantization results for DeiT-Tiny on ImageNet, using a state-of-the-art QAT approach LSQ+. The accuracy encounters severe degradation when the bit-width decreases to 3-bit.

tization has been massively studied for Convolution Neural Networks (CNN), few attention has been drawn to the quantization of ViT. For quantization, there are two approaches: Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT). Although PTQ is efficient for deployment, it is hard to achieve acceptable performance in ViT. For example, [21] only pushes its limit of quantization to 6-bit without great performance degradation. In contrast, QAT simulates quantization noise during the finetuning stage and shows more robustness to low-bit quantization. Results on CNN-based models have shown QAT is able to achieve low-bit quantization with negligible performance drop. Considering performance and robustness, it is potential to use QAT for low-bit ViT quantization.

As shown in Fig. 1, existing QAT methods suffer from a great performance drop when the ViT is quantized to 3- or 4- bits of precision. This is different from LSQ [10], in which the low-bit quantized CNN can reach full-precision accuracy. For deeper understanding, we conduct experiments to explore the quantization robustness of different components in the transformer layers. As is shown in Fig. 2a, in MLP, the GELU [14] is sensitive to quantization and need to be assigned with quantization bit-widths care-

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† Corresponding author.